Data Governance, Compliance and Ethics

FAIRNESS, ACCOUNTABILITY, AND TRANSPARENCY OF ALGORITHMIC SYSTEMS (PART 2&3)



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Overview of course content (MSC)

1. Data Governance and Management (Weeks 1-3)
2. Privacy and Data Protection (Weeks 4-6)
3. Ethical Issues Pertaining to Data (Weeks 7, 8)
4. Fairness, Accountability, and Transparency of Algorithmic Systems (Weeks11-13)

Fairness, Accountability, and Transparency of Algorithmic Systems (Part 2)

WEEKS 10, 11 &12

Agenda

The meaning of fairness with respect to algorithmic systems. Techniques and models for fairness-aware data mining, information retrieval, recommendation, etc. Legal, social, and philosophical models of fairness. Specification of mathematical objectives with respect to fairness. Perceptions of algorithmic bias and unfairness. Interventions to mitigate biases in systems, or discourage biased behaviour from users.

The meaning of transparency with respect to algorithmic systems.. Explanations for algorithmic logic and outputs. Trade-offs between privacy and transparency. Tools and methodologies for conducting algorithm audits. Frameworks for conducting ethical and legal algorithm audits. Empirical results from algorithm audits. The meaning of accountability with respect to algorithmic systems. Processes and strategies for developing accountable systems

Recap – FAT in ML

Fairness, Accountability and Transparency – way of examining critically one of the principal concerns about the use of algorithms in society (as part of the broader socio-technical system)

"In principle, **fairness** is the absence of any bias based on an individual's inherent or acquired characteristics that are irrelevant in the particular context of decision-making."

- To assess fairness, we need transparency
- To enforce accountability, we need transparency

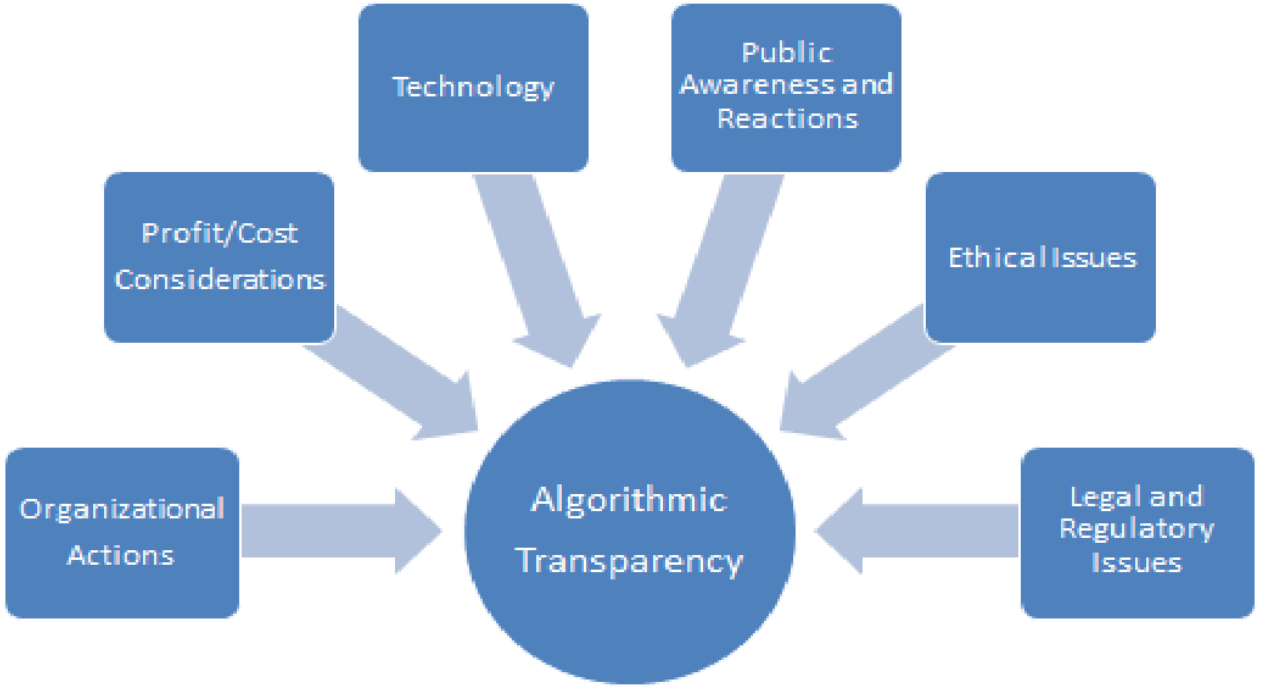


Algorithmic transparency

If an algorithm can be defined as **a set of steps** that a computer program follows in order to make a decision about a particular course of action, then

its transparency refers to the degree of openness about these steps, their purpose, structure and underlying actions of the **algorithms** that are used to search for, process and deliver information.

Of course, defining an algorithm is part of a socio-technical system and therefore broader than this technological approach and must be widened to included a range of other factors inside and outside the organisation using it.



Transparency can mean a range of different things

Note that 'transparency' has different meanings depending on who, what, why, how questions:

- access upon request to the public or authorised people;
- public posting of information;
- direct inspection of internal processes;
- the results of the manufacturer's or operator's tests of the system for accuracy and fairness;
- delivery of complete subsystems and their data for testing by authorised people, with the results reported to the public or to regulatory bodies;
- access to computer scientists and managers to explain algorithmic or operational processes.

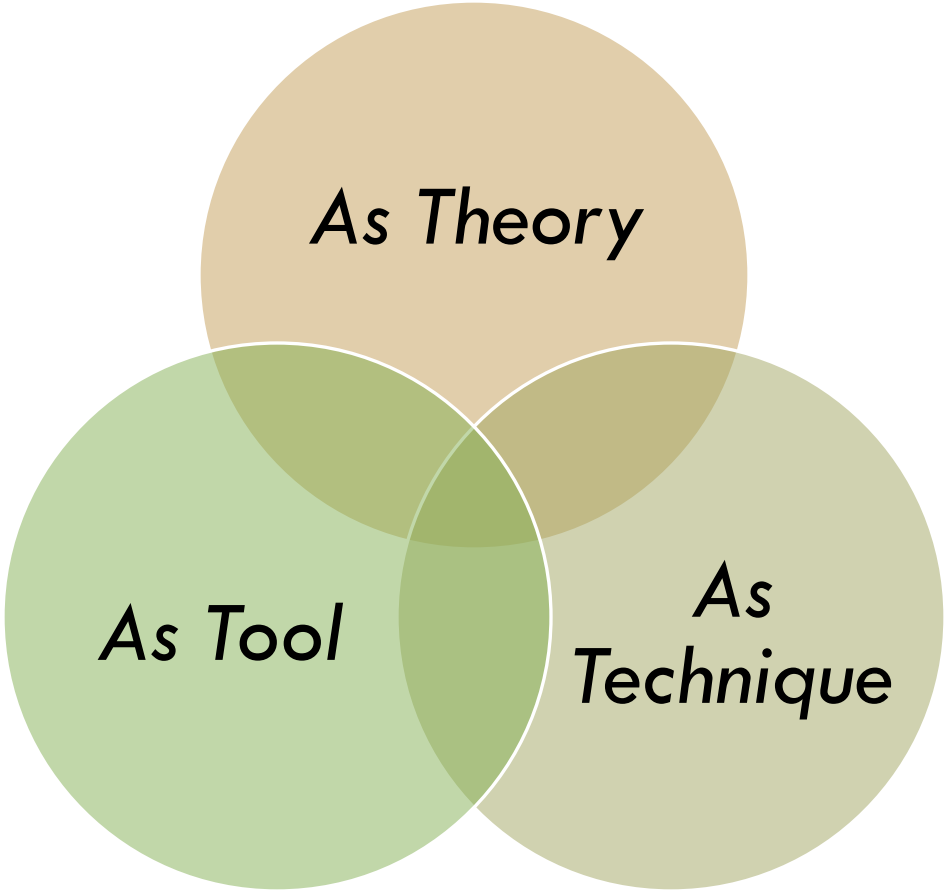
Source EPRS 2019

Transparency as theory, tool and technique

Theory/concept – relationship to fairness and accountability; how it underpins both

Tools – who, what, why?

Technique – practical methods to implement



Transparency as a tool to answer a range of questions (what, how, who?)

In the context of ensuring responsible development and use of algorithmic systems such that they improve human rights and benefit society, **transparency then is a tool to uncover the fairness properties of algorithms**.

Tools have three dimensions that are directly relevant to the use of transparency as a mechanism of governance of algorithmic systems:

1. **What?** A tool is valuable not in itself but because **of the goals its serves**; a can-opener is only useful if there are cans to be opened.
2. **Why/purpose?** No tool is right for every job. Misusing a tool has costs. Even using it appropriately often requires trade-offs.
3. **Who/whom?** the motivation of the person who **uses the tool** and what their ultimate objective is can be an important factor in its effectiveness.

Transparency of what?

Transparency is implied by the most basic conception of accountability: if we cannot know what an organisation is doing, we cannot hold it accountable, and cannot regulate it.

However we need to move beyond the concept of just knowing what is in the "black box" in order to ensure the kind of transparency that ensures accountability and ultimately the lack of bias an fairness

There are seven broad areas of machine learning systems about which transparency might be demanded

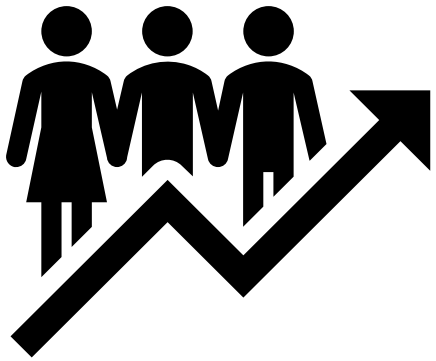
1. Data
2. Algorithms
3. Goals
4. Outcomes
5. Compliance
6. Influence
7. Usage

Source: European Parliamentary Research Service (2019)

Transparency of what? (1. Data)

The transparency of the data used by the algorithmic system -- in particular by machine learning and deep learning algorithms -- can refer to what knowledge we require of

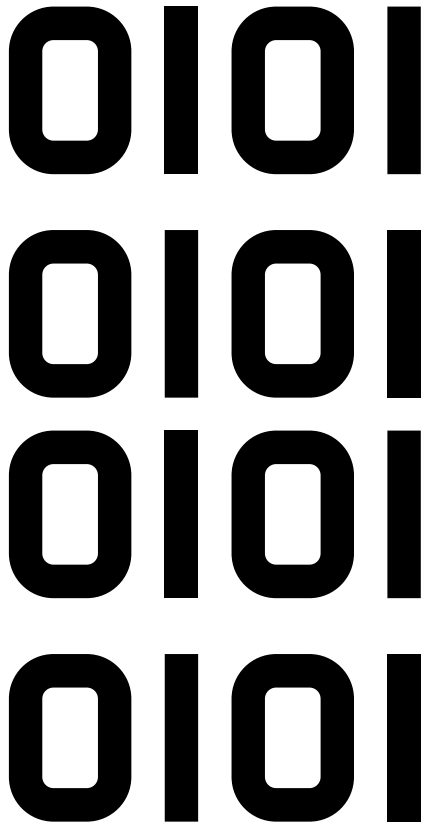
- the raw data,
- the data sources,
- how the data were pre-processed,
- the methods by which it was verified as unbiased and representative (including looking for features that are proxies for information about protected classes),
- the processes by which the data are updated and the system is trained on them.



Transparency of what? (2. Algorithms)

The transparency of the systems algorithms can refer to a number of approaches

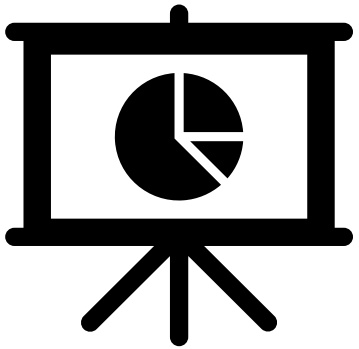
- How its output was tested against inputs (for which we know the proper output),
- How it reduced the variables to the most significant so we can validate them,
- How it tested the system (eg with counterfactuals) to see if prejudicial data is infecting the output,
- Whether a third-party code review, analysis of how the algorithms work, inspection of internal and external bug reports, or assurance the software development processes are sound.



Transparency of what? (3. Goals)

Transparency of Algorithmic systems can also refer to its goal or goals.

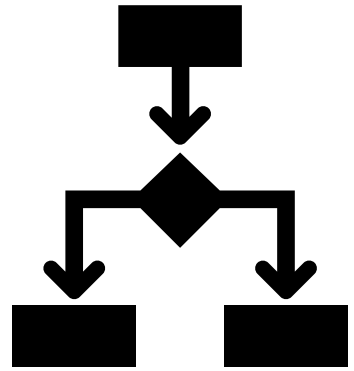
- What is it aiming to do? (eg grade/categorise, maximise, optimize, minimize)
- When a system has multiple goals, this would mean being transparent about their relative priorities.
- For example, the AI driving autonomous vehicles (AVs) might be aimed at reducing traffic fatalities, lowering the AVs' environmental impact, reducing serious injuries, shortening transit times, avoiding property damage, and providing a comfortable ride. What is the relative importance?
- A manufacturer could be required to be transparent about those goals and their priority.



Transparency of what? (4. Outcomes)

Developers of algorithmic systems could be required to be transparent about

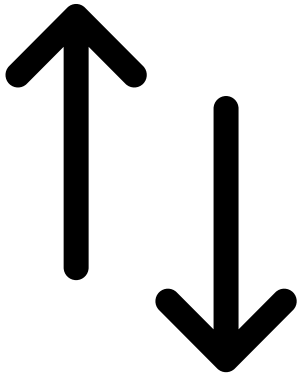
- the outcomes of the deployment of their algorithmic systems, including the internal states of the system (how worn are the brakes of an AV? how much electricity used?),
- the effects on external systems (how many accidents, or times it's caused another AV to swerve?), and
- computer-based interactions with other algorithmic systems (what communications with other AVs, what data fed into traffic monitoring systems?).



Transparency of what? (5. Compliance)

Manufacturers or operators may be required to be transparent about their overall compliance with whatever transparency requirements have been imposed upon them.

In many instances, we may insist that these compliance reports are backed by data that is inspectable by regulators or the general public.



Compliance with the transparency of data collection is at the heart of the GDPR. Data subjects need to be informed that their data is being collected and for what purpose.

GDPR recital 71 also provides for transparency in the case of "**appropriate mathematical or statistical procedures for the profiling** (of the data subject)". How is compliance to be shown in practice?

Transparency of what? (6. Influence)

Just as the public has an interest in knowing if an article in a newspaper was in fact paid for by an interested party, the public may have an interest in knowing if any element of the AI process was purposefully bent to favour a particular outcome.



For example, if a trusted search platform is artificially boosting some results because they were paid to, and if it is not flagging that fact to users, users can be manipulated.

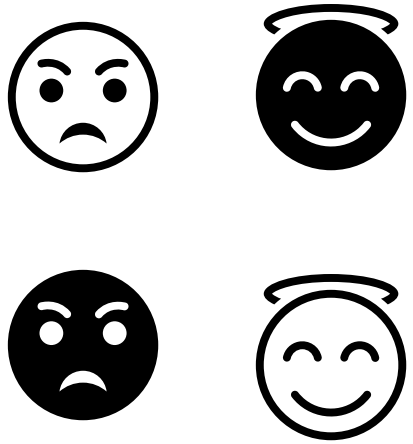
Regulators might want to insist that such influence be conspicuously acknowledged.

Transparency of what? (7. Usage)

Users may want to know what personal data a system is using, either to personalise outcomes or as data that can train the system to refine it or update it.

Knowing what personal data is used, they may then want to control that usage, perhaps to make their personalised results more accurate, or, more urgently, because they feel that usage violates their privacy, even though the data in question may already be a desired part of the system, such as a purchase or search history.

There are grey areas here as well: collecting anonymised, highly detailed information about trips made by autonomous vehicles — how often the car brakes or swerves, for example — could be important to optimizing traffic for safety or fuel efficiency.



Transparency - why?

We want systems to be transparent not to satisfy idle curiosity, but to help achieve important social goals related to accountability.



In effect, we want transparency of algorithmic system's data and algorithms in order to:

- Check for bias or unfairness
- Data relevance and representativeness.
- Model accuracy
- Model robustness and freedom from bugs
- Model integrity (Guard against malicious/adversarial data injection)



Transparency - why?

We want the **hierarchy of goals and outcomes** to be transparent so:

- It can be debated and possibly regulated.
- Regulators and the public can assess how well an algorithmic system has performed relative to its goals and compared to the pre-algorithmic systems it may be replacing or supplementing.

We want an **organisation's compliance status** to be public so:

- Regulators can hold the organisation accountable in case of failure.
- The public can evaluate the trustworthiness of the organisation, so people can make informed decisions as users about the services offered, and so citizens can become better informed about the benefits, risks, and trade-offs of algorithmic-based services overall.

Transparency by or for whom? (1/3)

Because transparency often, if not always, has costs and risks, it matters **who** gets to see what is illuminated. When considering regulating transparency the potential viewers include:

- Everyone: fully open access to data, algorithms, outcomes, etc.
- Regulatory authorities .
- Third-party forensic analysts whose reports are made public, made selectively public, or kept private.
- Researchers, possibly limited to those affiliated with accredited organisations and/or funding bodies.

Trade-offs between privacy and transparency (2/3)

Search platforms tend to give little information about exactly what criteria their algorithms use, at least in part because doing so would enable commercial interests to 'game' the system, making the results ranking less useful and reliable.

Therefore, algorithmic transparency can be at odds with the public (and commercial) interest in producing reliable, accurate search results.

This can be addressed by keeping the algorithmic inspection limited to trusted experts who are not permitted to disclose what they learn. This of course also has some risks: disclosure by accident or corruption.

Another risk is that Data Privacy could be compromised.

- The data used to train a model is typically similar to that used by the model -- and in cases where this is data about individuals, the training data may be protected. While this would preclude releasing the data itself (and disclosing the learning algorithm without the data provides little transparency into actual decisions), it also poses problems with releasing the model.
- It may be possible to 'reverse engineer' a model to determine the data used to construct it, thus violating privacy. While there are methods to prevent this(e.g., differential privacy), using such privacy protection methods can have disadvantages, both in terms of accuracy (although the impact on accuracy is debatable), and in complexity of system development.

Transparency by or for whom? (3/3)

Some of these potential costs can be mitigated by choosing where and how transparency interventions are necessary.

- For example, rather than providing direct public access to the data being used to train a machine learning system, independent data scientists could examine the data in private and publish the conclusions of their forensic research.

Transparency is a means not an absolute objective and thus needs to be negotiated depending on its purpose and the balance of benefits and costs.

- if an individual believes that s/he has been discriminated against by a black-box algorithmic system, but there is no evidence of systematic bias, the system might be tested to see if discriminatory factors were determinative in that particular outcome.
- Such testing might not require transparency. For example, inputting counterfactual data — say, a loan application in which only a factor is changed at a time — can identify the impact of possibly prejudicial data without requiring full transparency.

Summary

Transparency is a tool to be used responsibly, which means accepting that applying it means being sensitive to the complex contexts in which it is used, and the balance of benefits and harms its use inevitably entails.

As with any tool, whether and how it should best be used depends upon the context:

- The **goals** of requiring transparency.
- Which **elements** of the process should be made transparent.
- What **type of transparency** is most beneficial and least costly.
- **Alternative ways** of achieving the goal.
- A judgement of the potential trade-off between risk and the benefit an AI system could bring compared with the system it is replacing or augmenting.



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Transparency Components

Human factors

Data factors

Model design

Model inferences

Model projections

Table 1: possible components of algorithmic transparency (taken from Tow Center (2015))

Category	Type of Data to disclose
Human Involvement	The **goal, purpose, and intent** of the algorithm **Who** is the developer, **who** has direct control over the algorithm, **who** has oversight and is accountable
Data	The **quality** of the data, including its accuracy, completeness, and uncertainty, as well as its timeliness, magnitude and assumptions Dimensions of **data processing** such as how data was collected, transformed, vetted, and edited Whether the data the algorithms uses is **private or public**
The Model	The **features** or **variables** used in the algorithm, as well as any **weighting** of these factors and the **rationale** for this weighting The **assumptions** (statistical or otherwise) behind the model
Inferencing	**Benchmarks** of the inferences used against standard datasets The margin of **error** and **accuracy** rate (number false positives & false negatives) Steps are taken to remediate known errors Range of those confidence values as a measure of **uncertainty** in the outcomes.
Personalisation	Types of **personal information** being used

Source: Algorithmic transparency and accountability
Authors: Niklas Kossow, Svea Windwehr and Matthew Jenkins

Auditing Algorithms

First, consider that algorithms can be manipulated in ways that do not disadvantage their users directly or obviously.

Second, algorithmic manipulation may be societally problematic and deserving of scrutiny even if it is not illegal.

- This is a significant observation because the majority of scholarship that has considered algorithmic discrimination in the past has done so from the perspective of law and regulation.

As algorithms (and computers) become more common in the implementation of all technological systems, studying the world means studying algorithms (auditing).

Auditing Algorithms



There are two modes of algorithmic audit:

Direct auditing consists of code audits and other more traditional efforts, which are effective on machine learning systems and models that human auditors can deconstruct and interpret. For example, common data science methods such as decision trees and regression models generally can be analyzed and understood at the code level. This makes isolating the impact of all the inputs and variables in the model much simpler and more transparent.

Indirect auditing entails feeding sets of data that vary widely into an algorithm to test the outputs for signs of bias, anomalous behavior or other undesirable results. Indirect examination is useful in any audit, but is essential in audits of deep learning systems that write code autonomously.

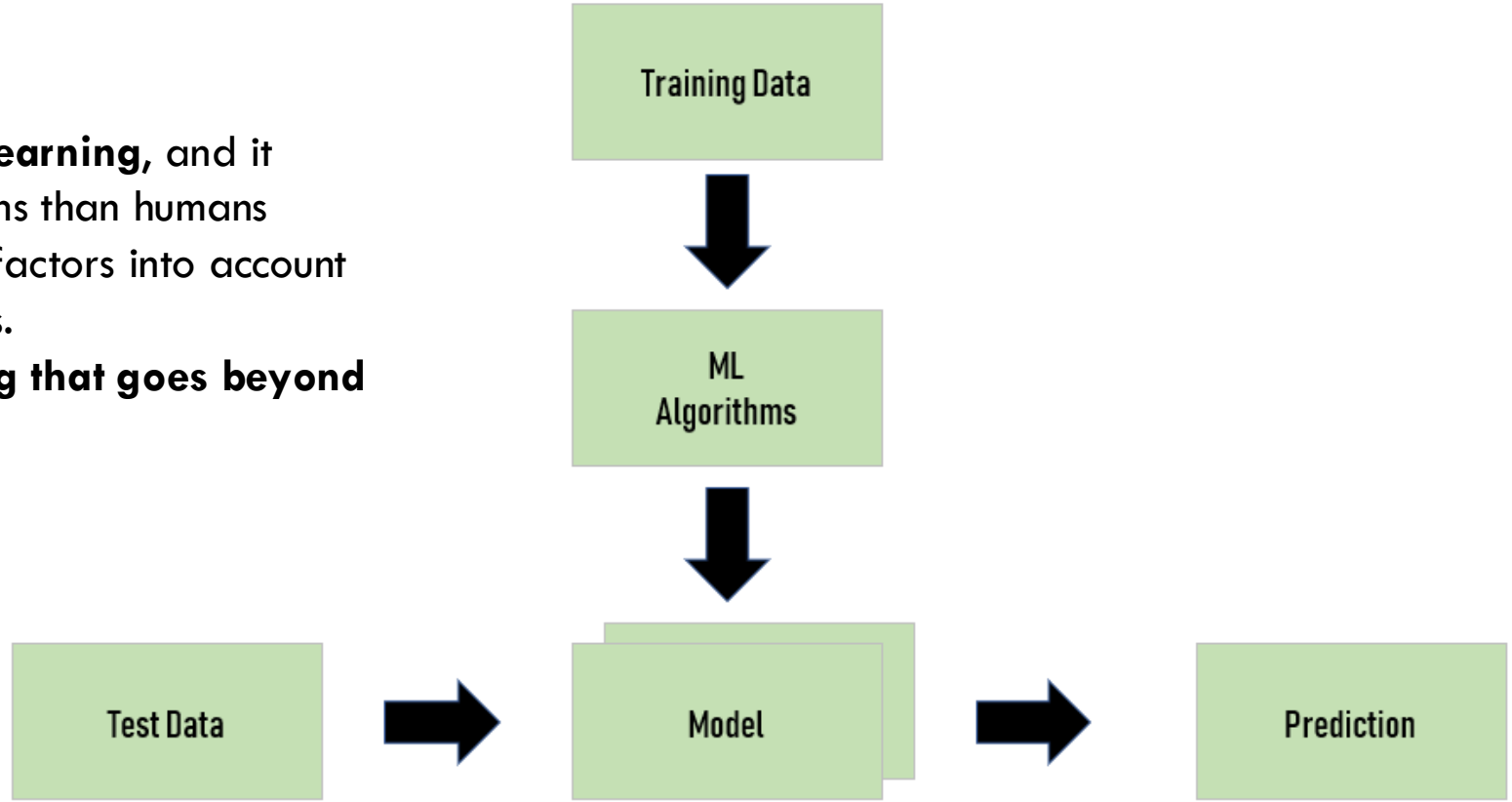
“Regular” vs Machine learning Algorithms

We can consider that there are two types of algorithm:

- a “regular” algorithm.
  - ***Algorithm:*** *process or set of rules to be followed in calculations or other problem-solving operations, especially by a computer.*
- a “Machine Learning” Algorithm.
  - ***Machine learning****: set of algorithms that enable the software to update and “learn” from previous outcomes without the need for programmer intervention. It is fed with structured data in order to complete a task without being programmed how to do so.*

“Regular” vs Machine learning Algorithms

ML is also called **model-based learning,** and it allows AI to make better decisions than humans because it can take many more factors into account and analyze them in milliseconds.
By using ML, you get something that goes beyond the sum of its parts.



“Regular” vs Machine learning Algorithms

An ML solution is truly based on Artificial Intelligence (AI) if it is not programmed **to perform a task, but is programmed to learn to perform the task**

ML algorithms usually require lesser computation and lesser data to perform reasonably well on many problems which are perfect for some business issues.

In classical algorithms, statisticians emphasis on p-value more and a solid but comprehensible model

Traditional learning methodologies such as training a model-based on historic training data and evaluating the resulting model against incoming data are not feasible as the environment is always changing

A traditional algorithm outputs results, whereas an ML algorithm outputs a model that outputs results.

Auditing Algorithms

Algorithmic audits are new—so new that standardized methodologies don't yet exist. For now, researchers generally agree on a few basic guidelines for algorithmic audits.

Check your data

- The most common cause of algorithmic bias is biased data. Reducing that bias means understanding what data has been collected and its relevance to the problem you're trying to solve, the integrity and accuracy of the data and the process undertaken to clean the raw data.

Check how the algorithm works

- Understanding the components and weighting of an algorithm makes it significantly easier to perform an audit. The very act of asking for an explanation of how an algorithm works can spark change.
- In particular, experts recommend asking what steps developers have taken to ensure the model is not biased against different protected groups.

Auditing Algorithms

Run your own "sniff test"

- One simple way to check for bias without having to analyze the source code is to run some test data points through an algorithm and check the result. Running these simple "sniff tests" can often be useful in spotting the worst cases of bias quickly and affordably.

Request a full audit

- There are various levels of algorithmic audits. A company can hire an outside firm to come in and lead the effort. If the company has an internal data team that is developing data models, it can institute a basic auditing process that involves an extra layer of testing outcomes to look for signs of bias.

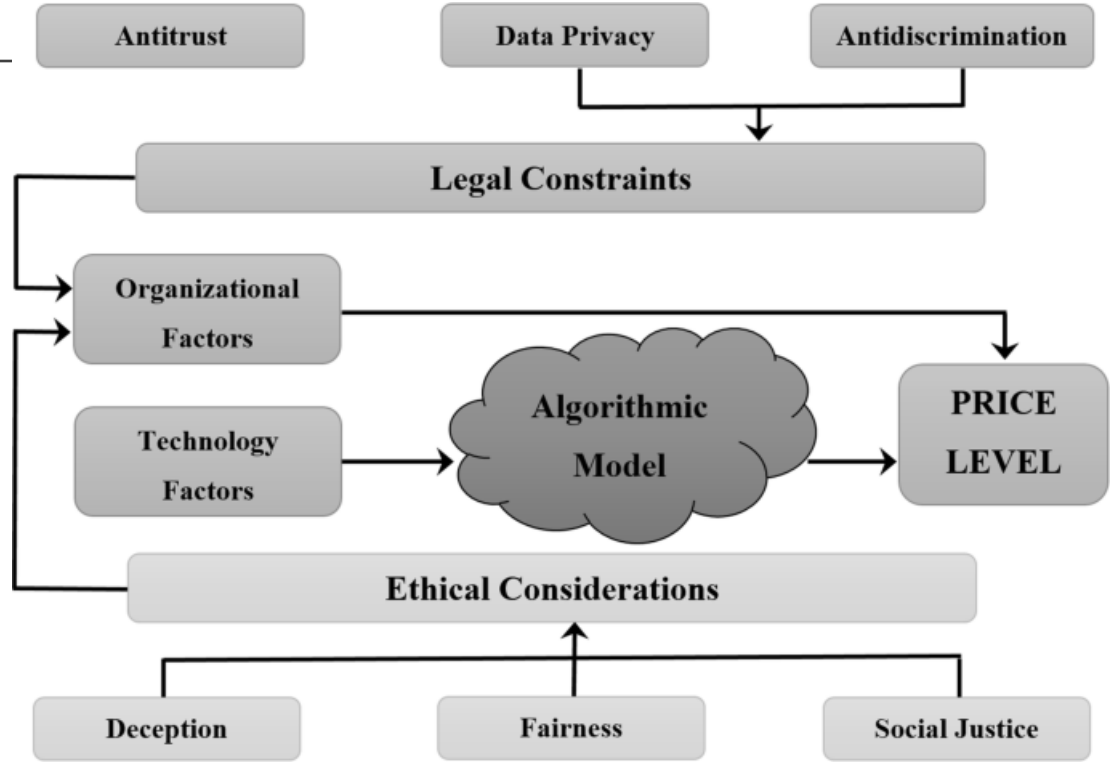
Audit regularly

- Bias—or just simply unfair results—may not be obvious initially and may only emerge as a model evolves or is trained on new data. Developers, who are used to writing code, checking it for errors, and then shipping it, aren't used to having to work this way.

Frameworks for conducting ethical and legal algorithm audits

Conceptual model of ethical and legal considerations of artificial intelligence

- Recognise an ethical issue
- Gather facts and evidence
- Evaluate alternative options
- Select option and evaluate/test it
- Reflect on outcome



https://www.scu.edu/media/ethics-center/ethical-decision-making/A-Framework-for-Ethical-Decision-Making.pdf

Benefits and risks from algorithm audits

Positive

- find biases
- keeping biases out of algorithms
- How might the algorithm be abused?'
- 'What harm might it cause on a portion of society?'

Negative

- exposing your algorithms may destroy your competitive advantage.
- for the people who are tech savvy, publishing the algorithm might allow them to understand the decision-making process in detail, thus allowing them effectively to game the system.

https://www.ft.com/content/879d86d6-93db-11e8-95f8-8640db9060a7

Tools for detecting Algorithmic Bias in AI

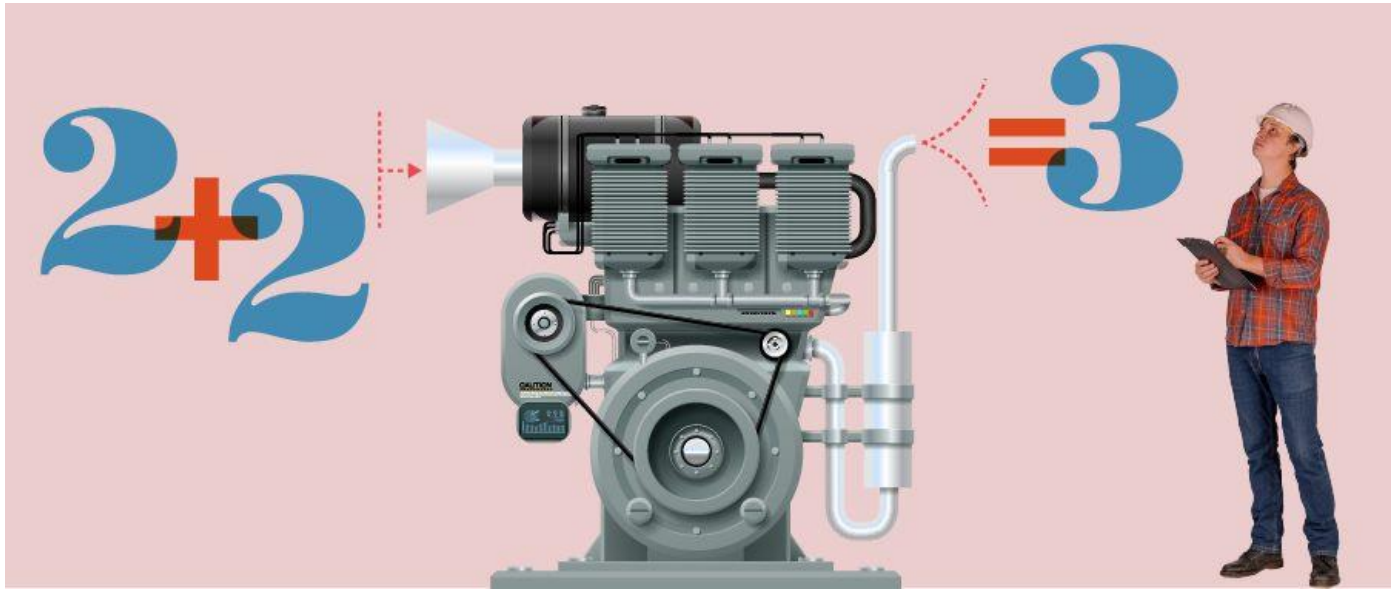
Pymetrics: Audit AI

DataScience.com Labs: Skater

Google: What-If Tool

IBM: AI Fairness 360 Open Source Toolkit

Accenture: Teach & Test AI Framework



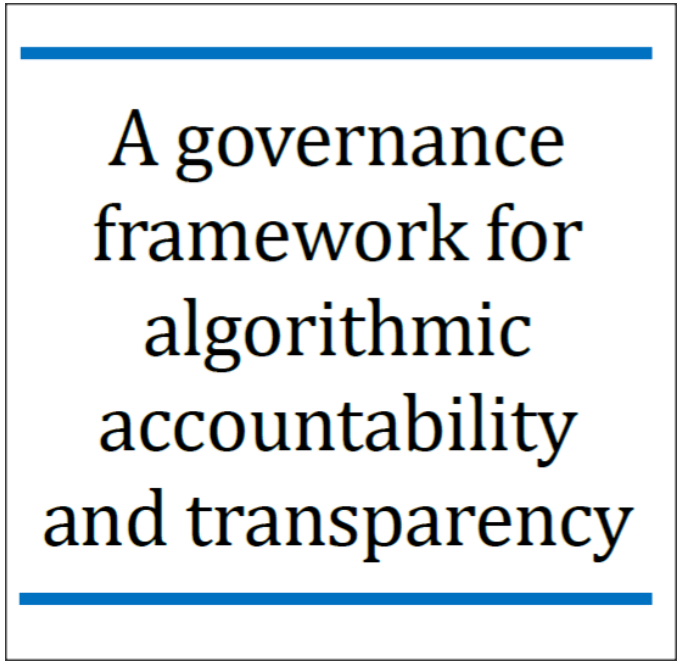
Algorithmic Accountability

PRINCIPLES FOR ACCOUNTABLE ALGORITHMS

Defining accountability

Accountability - 'A set of mechanisms, practices and attributes that sum to a governance structure which involves committing to legal and ethical obligations, policies, procedures and mechanism, explaining and demonstrating ethical implementation to internal and external stakeholders and remedying any failure to act properly'.

Source: EPRS (2019)





Meaning and functions of accountability

Accountability is primarily a legal and ethical obligation on an individual or organisation to account for its activities, accept responsibility for them, and to disclose the results in a transparent manner

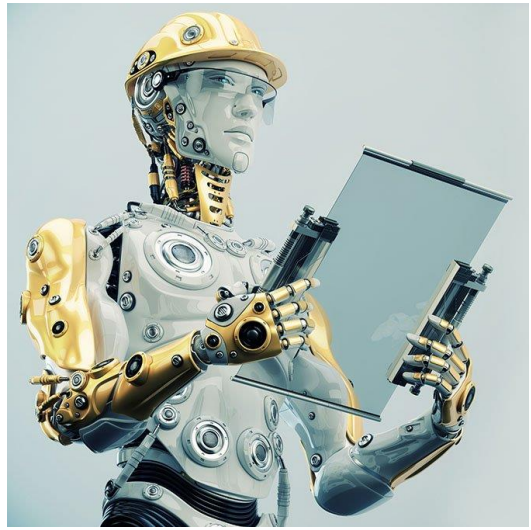


Functions of accountability principle

- act as a deterrent to reckless, irresponsible or illegal behaviour on the part of humans deploying/using algorithmic systems
- generate a self-reflective feedback loop for citizens and society, exposing existing biases and power dynamics

Algorithmic accountability

- "**Algorithmic accountability** ultimately refers to the assignment of responsibility for how an algorithm is created and its impact on society; if harm occurs, accountable systems include a mechanism for redress."
- "Importantly, the final decisions to put an algorithmic system on the market belongs to the technology's designers and company. **Critically, algorithms do not make mistakes, humans do**"
- An implication of this is that if harm occurs, accountable systems include a mechanism for redress. Especially in cases of technological redlining (application refusal), **assigning responsibility** is critical for quickly remediating discrimination and assuring the public that proper oversight is in place.
- In addition to clearly assigning responsibility, accountability must be grounded in enforceable policies that begin with auditing in pre- and post- marketing trials as well as standardized assessments for any potential harms.



Challenges for algorithmic accountability (1/2)

1. **Complex interactions between sub-systems and data sources**, some of which might not be under the control of the same entity (e.g., systems relying on data acquired through data brokers who rely on data sources that use algorithmic inference to aggregate over 'similar' data subjects).
2. **Unexpected outcomes associated with the impossibility of testing** against all possible input conditions when there are no methods for generating formal proofs for the system's performance.
3. **Difficulties in translating algorithmically derived concepts** (e.g. clustering algorithm results that segment populations based on large numbers of input variables) into human understandable concepts (e.g. ethnic affiliation) resulting in incorrect interpretations of the meaning of algorithmic results.

From: A governance framework for algorithmic accountability and transparency

Challenges for algorithmic accountability (2/2)

4. **Information asymmetries** arising from algorithmic inferences and black box processes that make it all but impossible for data subjects to gauge which, potentially false, information might have resulted in a particular algorithmic outcome affecting them (including lack of knowledge that algorithmic processes were even involved).
5. **Ubiquity of (small) algorithmic decisions which, if systematically biased,** may accumulate to have significant impacts on people even though no single decision would have achieved that legal threshold (e.g., impact on personal development due to reinforcement of racial/gender stereotypes by algorithmic recommendations).
6. **Purposeful injections of adversarial data** to fool a system into making errors, often in ways that can be very difficult to detect.

Principal of accountability is contested

"Currently, it is difficult to get technology corporations to answer for the harms their products have caused." (Caplan et al., 2018)

Why should this be?

- Developers argue that algorithms are objective, neutral, blank-slates leaving minimal responsibility for the developer. It is the responsibility of users to interpret the results correctly.
- Users on the other other hand argue that algorithms are black boxes, complicated, difficult to explain.

Algorithms have life-changing implications and therefore moral consequences? Who should be accountable for them particularly for redress if they go wrong?

Need for accountable algorithmic development (1/3)

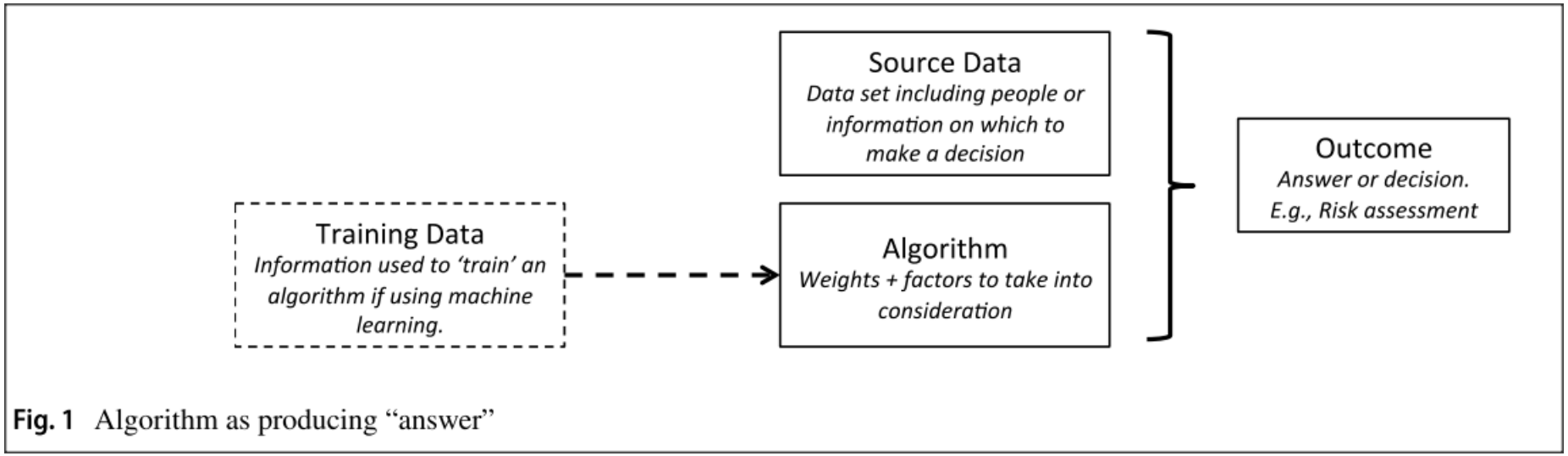


Fig. 1 Algorithm as producing "answer"

Martin, K. (2019) 'Ethical Implications and Accountability of Algorithms', Journal of Business Ethics, 160(4), pp. 835–850. doi: 10.1007/s10551-018-3921-3.

Need for accountable algorithmic development(2/3)

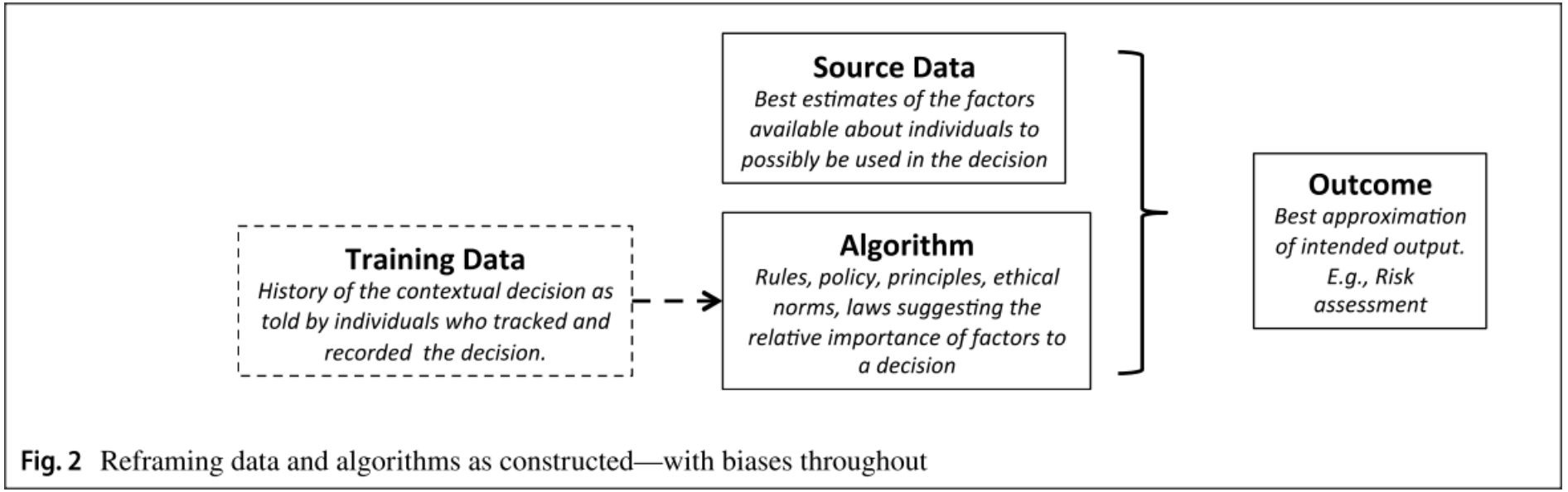


Fig. 2 Reframing data and algorithms as constructed—with biases throughout

Martin, K. (2019) 'Ethical Implications and Accountability of Algorithms', Journal of Business Ethics, 160(4), pp. 835–850. doi: 10.1007/s10551-018-3921-3.

Need for accountable algorithmic development(3/3)

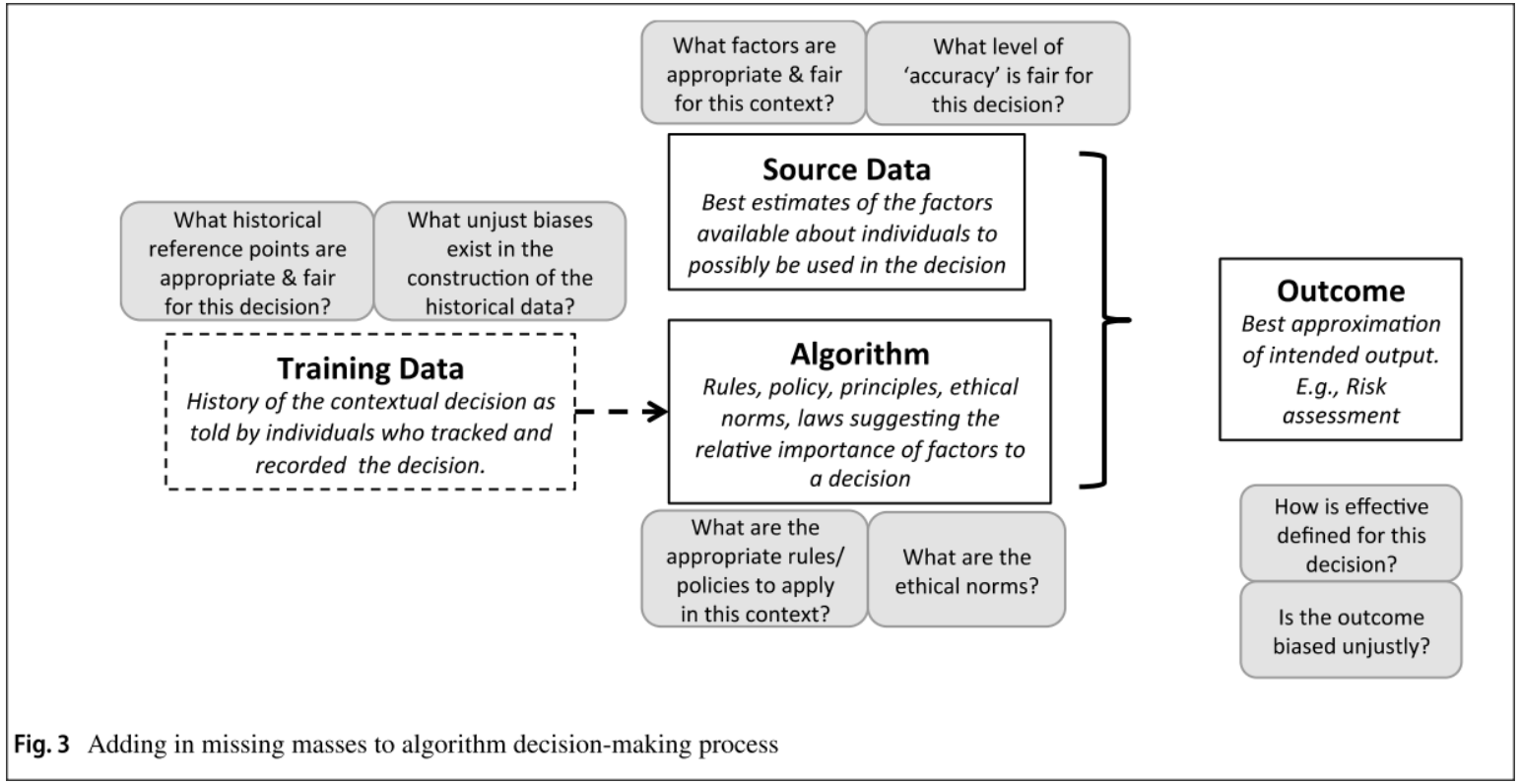


Fig. 3 Adding in missing masses to algorithm decision-making process

Martin, K. (2019) 'Ethical Implications and Accountability of Algorithms', Journal of Business Ethics, 160(4), pp. 835–850. doi: 10.1007/s10551-018-3921-3.

Framework for appropriate level corporate accountability (1/2)

Martin (2018) then proposes that the level of corporate accountability depends on two decisions that contributes to the appropriate type of accountability expected

- **Role of the algorithm in a decision** (e.g is it deciding to target an advertisement at us or is it being using as the primary basis for granting us a mortgage?) [**y-axis**]
- **Significance of the decision in societal terms** (e.g., is it suggesting a film to watch or recommending whether we should get released from prison?) [x-axis]

A range of circumstances there for will determine what the responsibility and accountability of a firm who developed the algorithm. This will depend upon (1)whether the algorithm plays a large or small role in the decision (2) the context in which the algorithm is to be used .

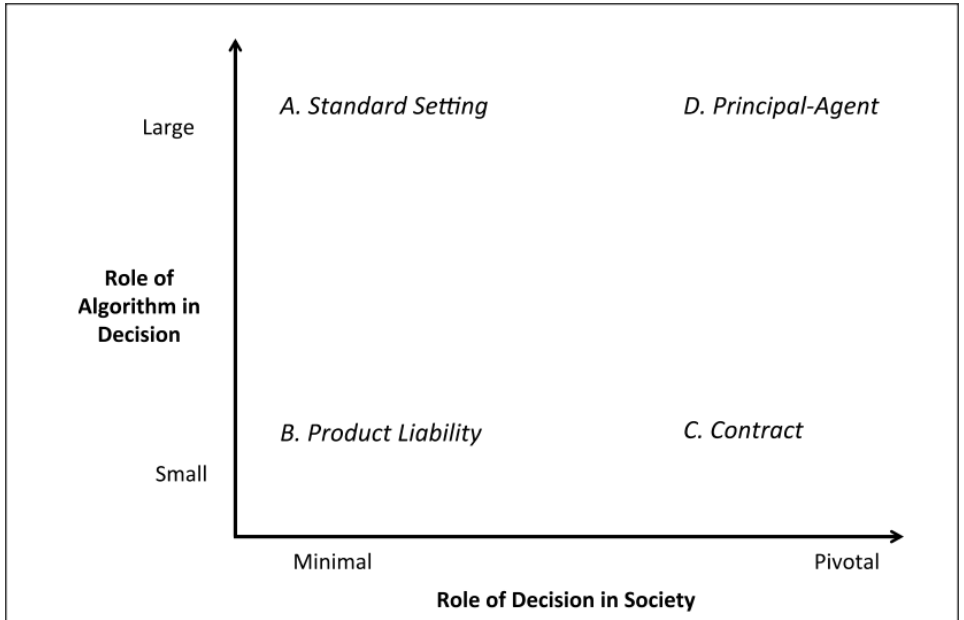


Fig. 4 Firm responsibility for algorithms

Framework for appropriate level corporate accountability (2/2)

A. **Standard setting:** This is where a developer firm acts as an expert and designs an algorithm to take a larger role in the decision of minimal societal importance such as where to place an ad online as well as deciding upon the delegation of roles and responsibilities encoded in the design

B. **Product liability:** In this situation where the algorithm plays only a minor role in a decision of minor societal importance, the role of the developer is of a traditional product liability with minimal responsibility

C. **Contract:** Where the algorithm plays a smaller role in a significant decision such as sentencing, this relationship his closer to a contract with the responsibility to remain engaged well the algorithm is used in case the role changes.

D. **Principal-agent:** In a deployment where the decision provides a gatekeeping function to important social goods such as sentencing, allocation of medical care, access to education etc. the relationship is that where the algorithm acts as an agent for the designer.

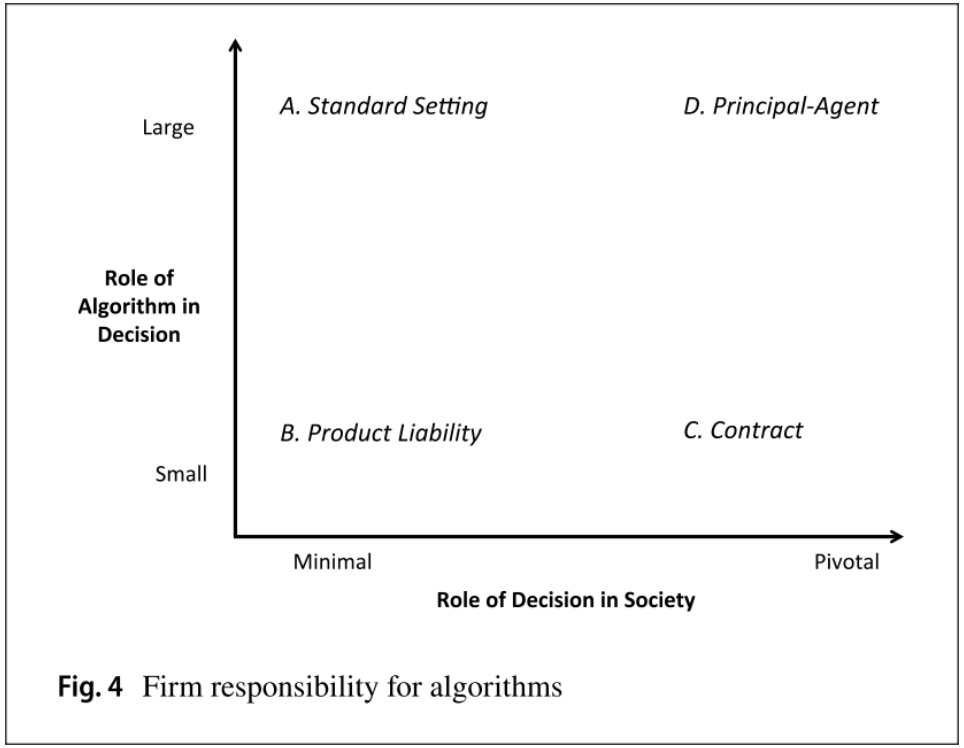


Fig. 4 Firm responsibility for algorithms

Sources and resources

Caplan, R. *et al.* (2018) *Algorithmic Accountability: A Primer*. Data&Society. Available at: https://datasociety.net/wp-content/uploads/2019/09/DandS_Algorithmic_Accountability.pdf.

Diakopoulos, N. *et al.* (2018) 'Principles for Accountable Algorithms and a Social Impact Statement for Algorithms :: FAT ML', *Fatml.Org*, pp. 1–5. Available at: https://www.fatml.org/resources/principles-for-accountable-algorithms (Accessed: 4 December 2020).

European Parliamentary Research Service (2019) *Scientific Foresight Unit (STOA), A governance framework for algorithmic accountability and transparency, Brussels: European Union 2019*. Available at: http://www.europarl.europa.eu/RegData/etudes/STUD/2019/624262/EPRS_STU(2019)624262_EN.pdf.

Kaminski, M. E. and Malgieri, G. (2019) 'Algorithmic Impact Assessments under the GDPR: Producing Multi-layered Explanations', *SSRN Electronic Journal*, pp. 1–29. doi: 10.2139/ssrn.3456224.

Martin, K. (2019) 'Ethical Implications and Accountability of Algorithms', *Journal of Business Ethics*, 160(4), pp. 835–850. doi: 10.1007/s10551-018-3921-3.

O'Keefe, K. and O'Brien, D. (2018) *Ethical data and information management : concepts, tools and methods*. KoganPage. (pages 88-91)